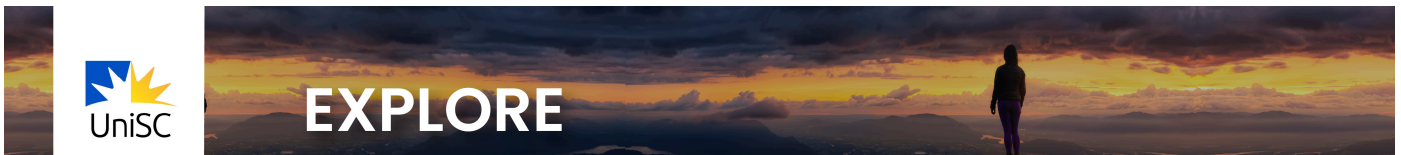


4.3 What is the security market line (SML)? How can it be used to see undervalued and overvalued stocks?



Guiding Questions

1. What is the Security Market Line (SML)?
2. If an asset's return is plotted above the SML, what is the interpretation?
3. If an asset's return is plotted below the SML, what is the interpretation?



What will you learn?

You will be introduced to the concept of SML. You will learn:

- Why SML is useful.
- Why any asset plotted above the SML is considered underpriced.
- Why any asset plotted below the SML is considered overpriced.

What is the Security Market Line?

Security Market Line (SML) is a graphical representation of the CAPM. It is a plot that has expected return from a security on the y-axis and its beta on the x-axis.

To develop an understanding of SML, consult the resources available [here](https://www.wallstreetprep.com/knowledge/security-market-line-sml/) 
(<https://www.wallstreetprep.com/knowledge/security-market-line-sml/>).



Watch

Security Market Line



Watch

Security Market Line: Overvalued or Undervalued

SML and Asset Pricing

Finance professionals use SML to spot assets that are mispriced relative to the CAPM prediction.

Graphically:

- 1) If an asset's return is higher than what the CAPM predicts for its level of systematic risk (as measured by its beta), it will be plotted above the SML. Such an asset is under-priced relative to the CAPM prediction.
- 2) If an asset offers a return lower than what the CAPM predicts for its level of beta, it will be plotted below the SML. Such an asset is overpriced relative to the CAPM prediction.

The intuitive reason why an asset that is plotted above the SML is under-priced is that it is offering a return which is too large. As returns and prices are inversely related, it follows that its price is too low. Similarly, an asset that is plotted below the SML is overpriced because it is offering a return which is too low implying that its price is too high. Of course, the above discussion assumes that CAPM is the true model. Empirical evidence contradicts this assumption (see the next section). Nevertheless, SML remains a useful tool widely used in the industry.

Solve

Q1) If the risk-free rate is 5%, the market return is 11%, and a stock has a beta of 1.5, what is its required return?

Q2) A stock offers a 16.5% return and has a beta of 1.5, given a 14% required return (calculated above), is it over or under priced?

Q3) What does a beta of 0 signify on an SML graph?

Q4) The market risk premium represents the slope of the SML. True or False.

Answers available [here. \(https://learn.usc.edu.au/courses/35359/files/3282425?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3282425?wrap=1)

Further Exploration

To see, how the concept of SML builds on the concepts discussed earlier in this course, consult the free pdf chapter available [here \(https://learn.usc.edu.au/courses/35359/files/3228822?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3228822?wrap=1).